

CHAPTER 7

INTRODUCTION TO FUZZY LOGIC, CLASSICAL SETS AND FUZZY SETS

What is Fuzzy Logic

Fuzzy logic is a convenient way to map an input space to an output space.

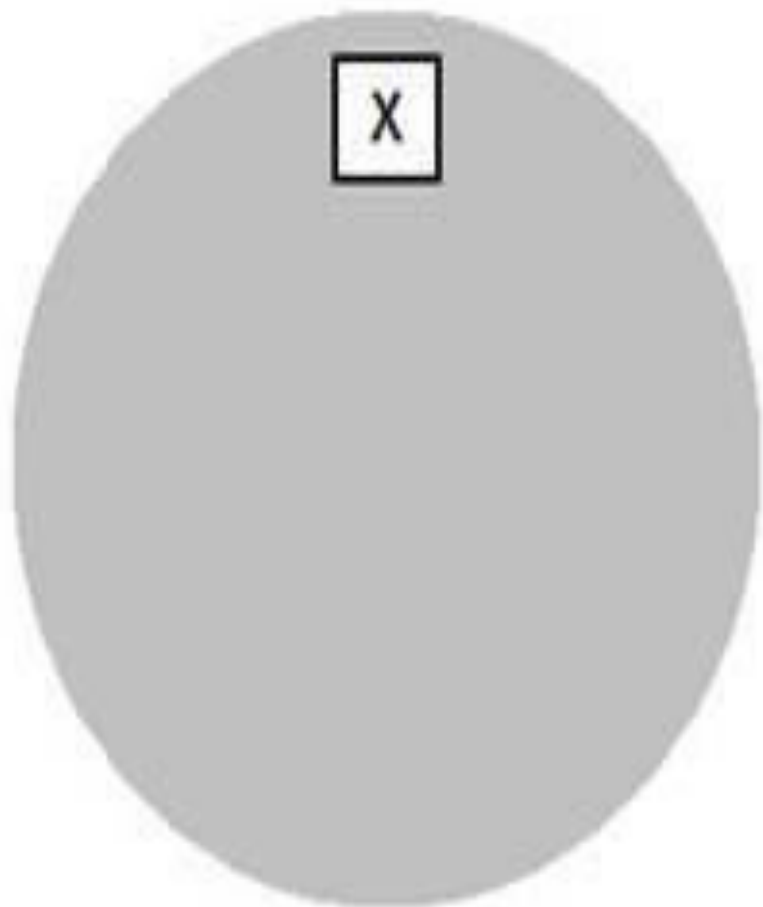
- * With information about how good your service was at a restaurant, a fuzzy logic system can tell you what the tip should be.
- * With your specification of how hot you want the water, a fuzzy logic system can adjust the faucet valve to the right setting.
- * With information about how far away the subject of your photograph is, a fuzzy logic system can focus the lens for you.
- * With information about how fast the car is going and how hard the motor is working, a fuzzy logic system can shift gears for you.

INTRODUCTION

- Fuzzy logic is an approach to computing based on "degrees of truth" rather than the usual "true or false" (1 or 0) [Boolean](#) logic on which the modern computer is based.
- Why Fuzzy Logic?
 - Fuzzy logic is useful for commercial and practical purposes.
 - It can control machines and consumer products.
 - It may not give accurate reasoning, but acceptable reasoning.
 - Fuzzy logic helps to deal with the uncertainty in engineering.

Not X

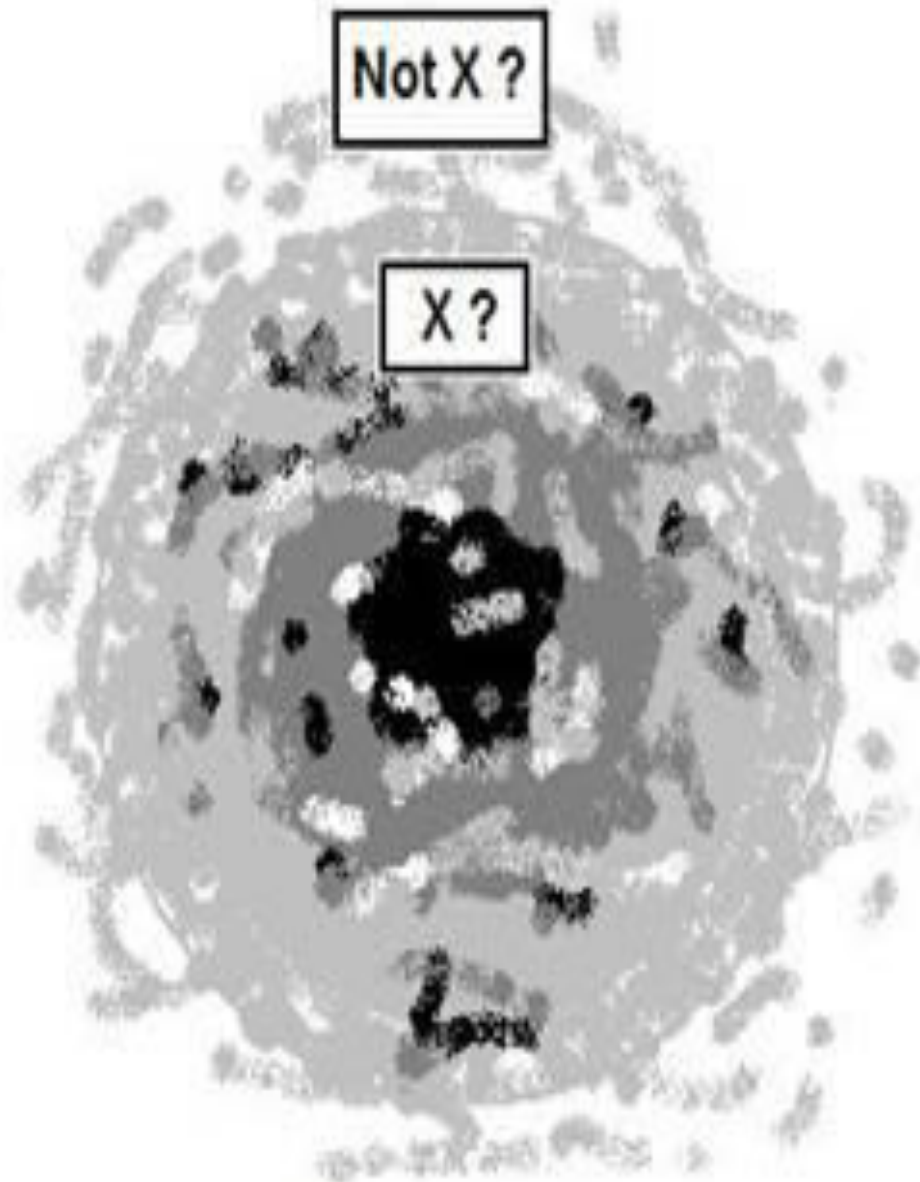
X



Definite logic

Not X ?

X ?



Fuzzy logic

Arsha J K, AP, Computer Science
and Engineering, VJCET,
Vazhakulam

Is Ram Honest?

True/Yes/1

False/No/0

Boolean Logic

Is Ram Honest?

Extremely Honest

Very Honest (0.85)

Sometimes Honest (0.35)

Extremely Dishonest (0.0)

Fuzzy Logic

FUZZY LOGIC REPRESENTATION

- For every problem must represent in terms of fuzzy sets.



Slowest
[0.0 – 0.25]



Slow
[0.25 – 0.50]



Fast
[0.50 – 0.75]



Fastest
[0.75 – 1.00]

- What are fuzzy sets?



WHY IS IT USEFUL?

Automatic Braking System

Fuzzy Logic



Is car close? : 0-1 (Range of No to Yes)

Brakes : 0-1 (Range of Off to On)

Arsha J K, AP, Computer Science
and Engineering, VJCET,
Vazhakulam

3. Transportation

Fuzzy logic controllers have been developed for automatic gearboxes in automobiles.

For example:

- Audi TT
- VW Toureg
- VW caravell

These features the DSP Transmission which utilizes fuzzy logic.

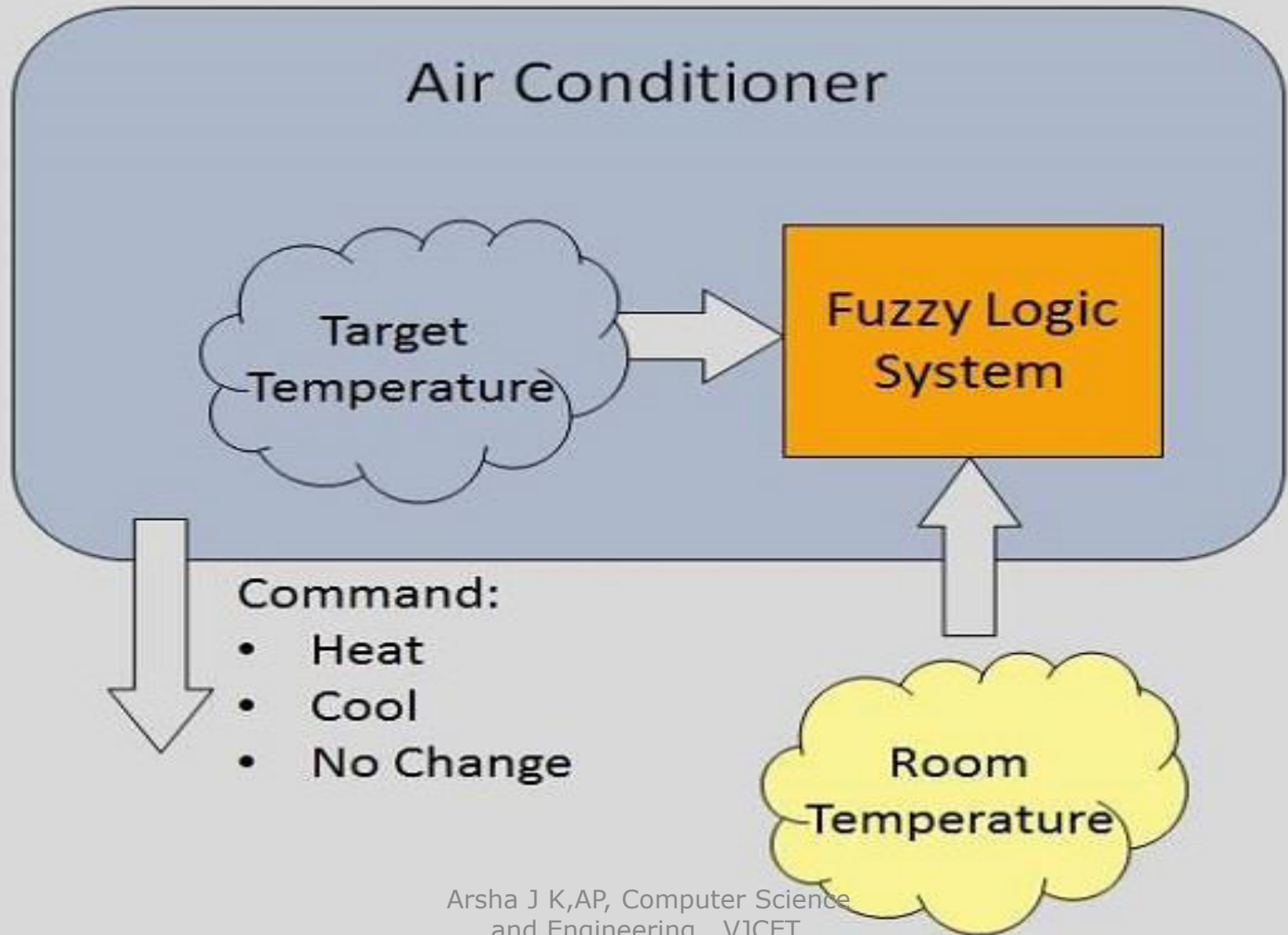


4. Toys And Games

AI has also been applied to video games, for example video game bots, which are designed to stand in as opponents where humans aren't available or desired.



Room



Fuzzy Logic in a Washing Machine



- **Fuzzy logic** washing machines are gaining popularity. These machines offer the advantages of **performance**, **productivity**, **simplicity**, **productivity**, and **less cost**. Sensors continually monitor varying conditions inside the machine and accordingly adjust operations for the best wash results. As there is no standard for fuzzy logic, different machines perform in different manners.

Why use Fuzzy Logic

"In almost every case you can build the same product without fuzzy logic, but fuzzy is **faster** and **cheaper**."

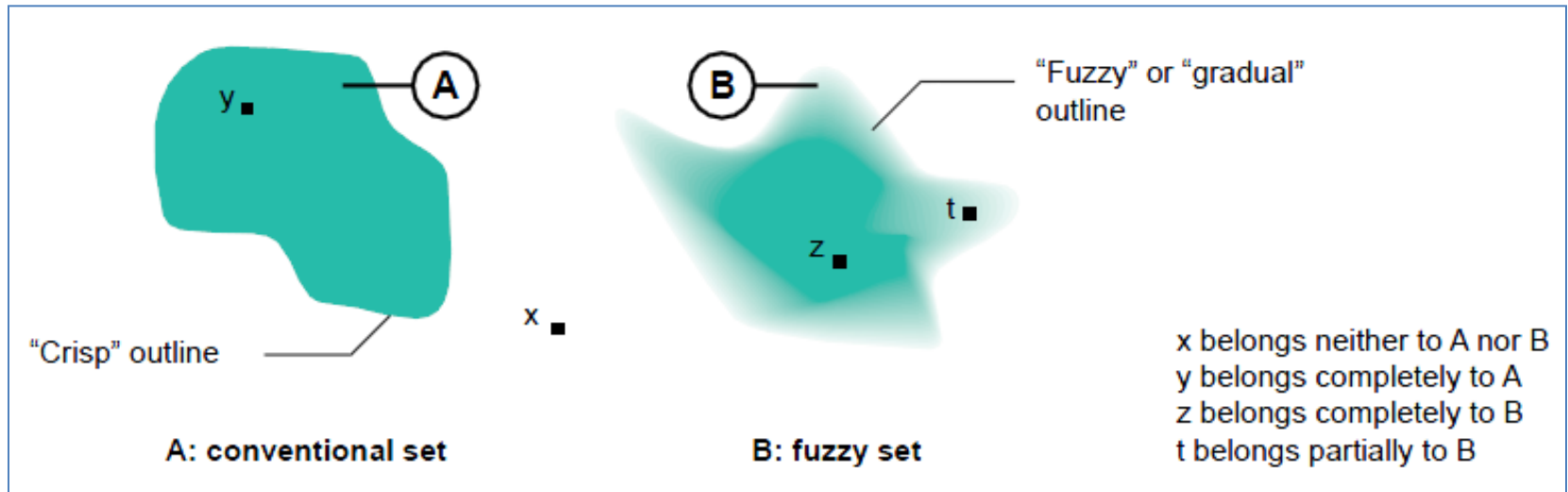
-Lotfi Zadeh

Why use Fuzzy Logic

Here is a list of general observations about fuzzy logic:

- Fuzzy logic is conceptually easy to understand.
- Fuzzy logic is flexible
- Fuzzy logic is tolerant of imprecise data.
- Fuzzy logic can model nonlinear functions of arbitrary complexity.
- Fuzzy logic can be blended with conventional control techniques.
- Fuzzy logic is based on natural language.

Classical Sets and Fuzzy Sets

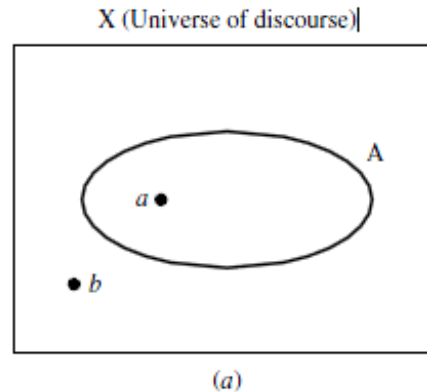


X (Universe of discourse)

A **classical set** is defined by *crisp boundaries*

A **fuzzy set** is prescribed by **vague** or **ambiguous** properties; hence its boundaries are ambiguously specified

Classical Sets



the universe of discourse is the universe of all available information on a given problem
a **universe of discourse**, X , as a collection of objects all having the same characteristics

The clock speeds of computer CPUs

The operating currents of an electronic motor

The operating temperature of a heat pump (in degrees Celsius)

The Richter magnitudes of an earthquake

The integers 1 to 10

The individual elements in the universe X will be denoted as x . *The features of the elements in X can be discrete, countable integers or continuous valued quantities on the real line.*

Arsha J K, AP, Computer Science

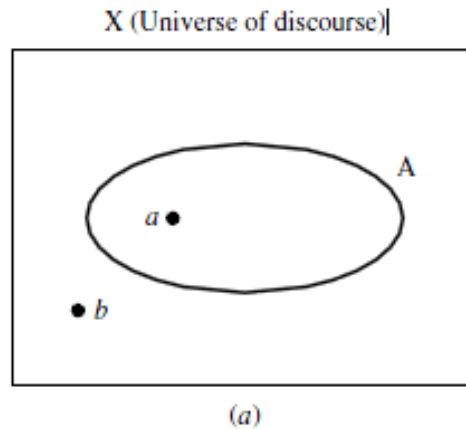
The total number of elements in a universe X is called its **cardinal number**, denoted n_x

and Engineering, VJGT
Vazhakulam

Classical Set/Crisp set

- Collection of objects which share same characteristics
- A classical set is the collection of distinct objects
- Consider an object in the crisp set A. This object is either the member or nonmember of the given set A
- In crisp set no partial membership exist
- Let Universe of discourse be U.
- The collection of elements in the universe is called whole set
- The total number of elements in the universe U is called cardinal number denoted by n_U
- Collection of elements within the universe is called set
- Collection of elements within the set is called subset

Classical Sets



Collections of elements within a universe are called **sets**

universe of discourse : The Richter magnitudes of an earthquake
Set in the universe of discourse?

Collections of elements within sets are called **subsets**

The collection of all possible sets in the universe is called the *whole set (power set)*.

Classical Sets

We have a universe comprised of three elements, $X = \{a, b, c\}$

The cardinal number, n_x ?

The power set, $P(X)$?

The cardinality of the power set?

CRISP SETS

- There are several ways for defining a set. A set may be defined using one of the following
 1. The list of all members of the set may be given. Example $A=\{2,3,4,5\}$
 2. The properties of the set element may be specified
Example $A=\{x \mid x \text{ is prime number } < 20\}$
 3. The formula for the definition of the set may be specified
 4. The set may be specified on the basis of the result of the logical operation
 5. There exist a membership function, which may also be used to define a set. The membership is denoted by the symbol μ and the membership function for the set A is given by

$$\mu_A(x) = \begin{cases} 1 & \text{If } x \text{ belongs to } A \\ 0 & \text{If } x \text{ not belongs to } A \end{cases}$$

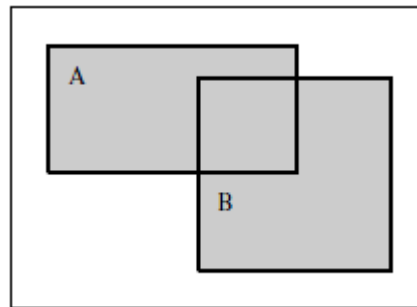
Operation on Classical Sets

Union

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

The **union** between the two sets, denoted $A \cup B$, represents all those elements in the universe that reside in (or belong to) the set A, the set B, or both sets A and B.

This operation is also called the **logical or**



Union of sets A and B (logical or) in terms of Venn diagrams

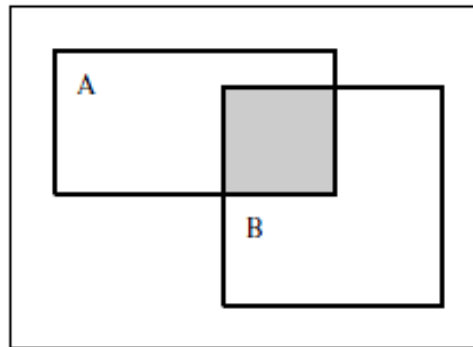
Arsha J K, AP, Computer Science
and Engineering, VJCET,
Vazhakulam

Operation on Classical Sets

Intersection $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

The **intersection** of the two sets, denoted $A \cap B$, represents all those elements in the universe X that **simultaneously** reside in (or belong to) both sets A and B .

This operation is also called the **logical and**



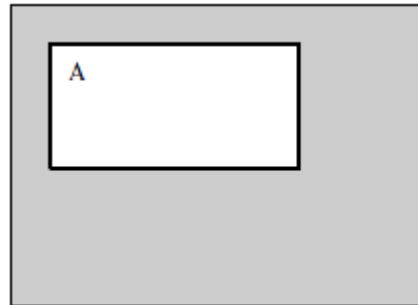
Intersection of sets A and B.

Operation on Classical Sets

Complement

$$\bar{A} = \{x \mid x \notin A, x \in X\}$$

The **complement** of a set A, is defined as the collection of all elements in the universe that do not reside in the set A.



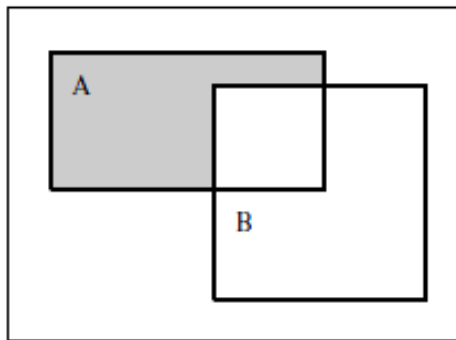
Complement of set A

Operation on Classical Sets

Difference

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$$

The **difference** of a set A with respect to B, denoted $A \setminus B$, is defined as the collection of all elements in the universe that reside in A and that do not reside in B simultaneously



Difference operation $A \setminus B$

Properties of Classical (Crisp) Sets

Commutativity $A \cup B = B \cup A$

$$A \cap B = B \cap A$$

Associativity $A \cup (B \cup C) = (A \cup B) \cup C$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributivity $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad (2.7)$$

Idempotency $A \cup A = A$

$$A \cap A = A$$

Identity $A \cup \emptyset = A$

$$A \cap X = A$$

$$A \cap \emptyset = \emptyset$$

$$A \cup X = X$$

Transitivity *If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$*

Properties of Classical (Crisp) Sets

Two special properties of set operations,

The *excluded middle* axioms
De Morgan's principles

Properties of Classical (Crisp) Sets

The *excluded middle axioms*

not valid for both classical sets and fuzzy sets.

There are **two** excluded middle axioms

The first, called the ***axiom of the excluded middle***, deals with the union of a set A and its complement,

the second, called ***the axiom of contradiction***, represents the intersection of a set A and its complement.

Axiom of the excluded middle

$$A \cup \bar{A} = X$$

Axiom of the contradiction

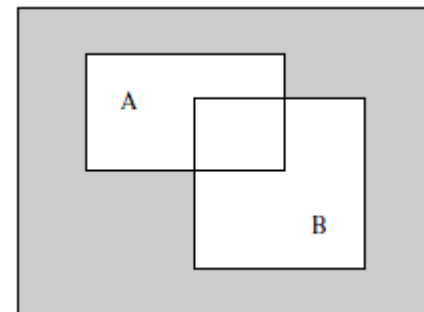
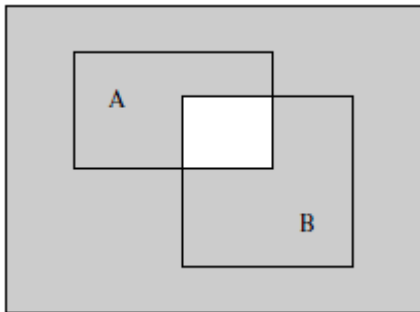
$$A \cap \bar{A} = \emptyset$$

Properties of Classical (Crisp) Sets

De Morgan's principles

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$



information about the complement of a set (or event), or the complement of combinations of sets (or events), rather than information about the sets themselves

Properties of Classical (Crisp) Sets

De Morgan's principles

Example: A shallow arch consists of two slender members as shown in Fig. If either member fails, then the arch will collapse.

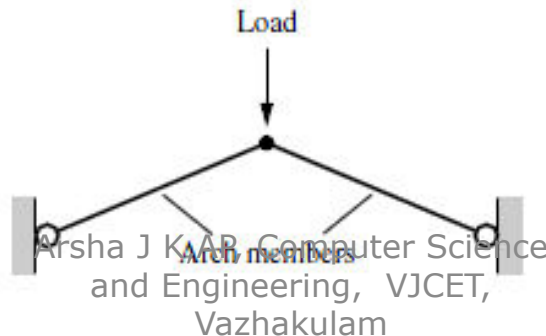
E1 = survival of member 1 and

E2 = survival of member 2,

Survival of the arch = ?

Collapse of the arch = ?

Logically, collapse of the arch will occur if either of the members fails



Properties of Classical (Crisp) Sets

De Morgan's principles

Example 1: A shallow arch consists of two slender members as shown in Fig. If either member fails, then the arch will collapse.

E_1 = survival of member 1 and

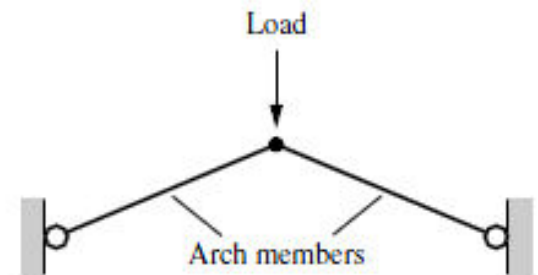
E_2 = survival of member 2,

Survival of the arch = $E_1 \cap E_2$

Collapse of the arch = $\overline{E_1 \cap E_2}$.

Collapse of the arch will occur if either of the members fails: $\overline{E_1} \cup \overline{E_2}$.

Illustration of De Morgan's principle: $\overline{E_1 \cap E_2} = \overline{E_1} \cup \overline{E_2}$



Mapping of Classical Sets to Functions

Mapping is an important concept in relating set-theoretic forms to function-theoretic representations of information.

In its most general form it can be used to map elements or subsets on one universe of discourse to elements or sets in another universe.

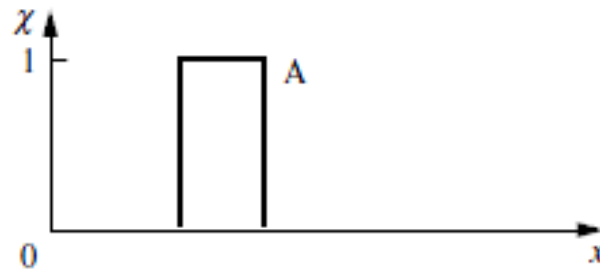
If an element x is contained in X and corresponds to an element y contained in Y , it is generally termed **a mapping from X to Y** ,

$$f: X \rightarrow Y$$

Mapping of Classical Sets to Functions

The characteristic (indicator) function χ_A is defined by

$$\chi_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$



Membership function is a mapping for crisp set A .

Mapping of Classical Sets to Functions

Example : a universe with three elements, $X = \{a, b, c\}$, we desire to map the elements of the power set of X , i.e., $P(X)$, to a universe, Y , consisting of only two elements (the characteristic function), $Y = \{0, 1\}$

the elements of the power set?

the elements in the value set $V(P(X))$?

Mapping of Classical Sets to Functions

Example : a universe with three elements, $X = \{a, b, c\}$, we desire to map the elements of the power set of X , i.e., $P(X)$, to a universe, Y , consisting of only two elements (the characteristic function), $Y = \{0, 1\}$

the elements of the power set

$$P(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$$

the elements in the value set $V(P(X))$

$$V\{P(X)\} = \{\{0, 0, 0\}, \{1, 0, 0\}, \{0, 1, 0\}, \{0, 0, 1\}, \{1, 1, 0\}, \{0, 1, 1\}, \{1, 0, 1\}, \{1, 1, 1\}\}$$

Mapping of Classical Sets to Functions

The **union** of these two sets in terms of **function-theoretic terms** is given as follows (the symbol $\vee$ is the maximum):

$$\text{Union } A \cup B \rightarrow \chi_{A \cup B}(x) = \chi_A(x) \vee \chi_B(x) = \max(\chi_A(x), \chi_B(x))$$

The **intersection** of these two sets in **function-theoretic terms** is given by (the symbol $\wedge$ is the minimum operator):

$$\text{Intersection } A \cap B \rightarrow \chi_{A \cap B}(x) = \chi_A(x) \wedge \chi_B(x) = \min(\chi_A(x), \chi_B(x))$$

The **complement** of a single set on universe X , say A , is given by

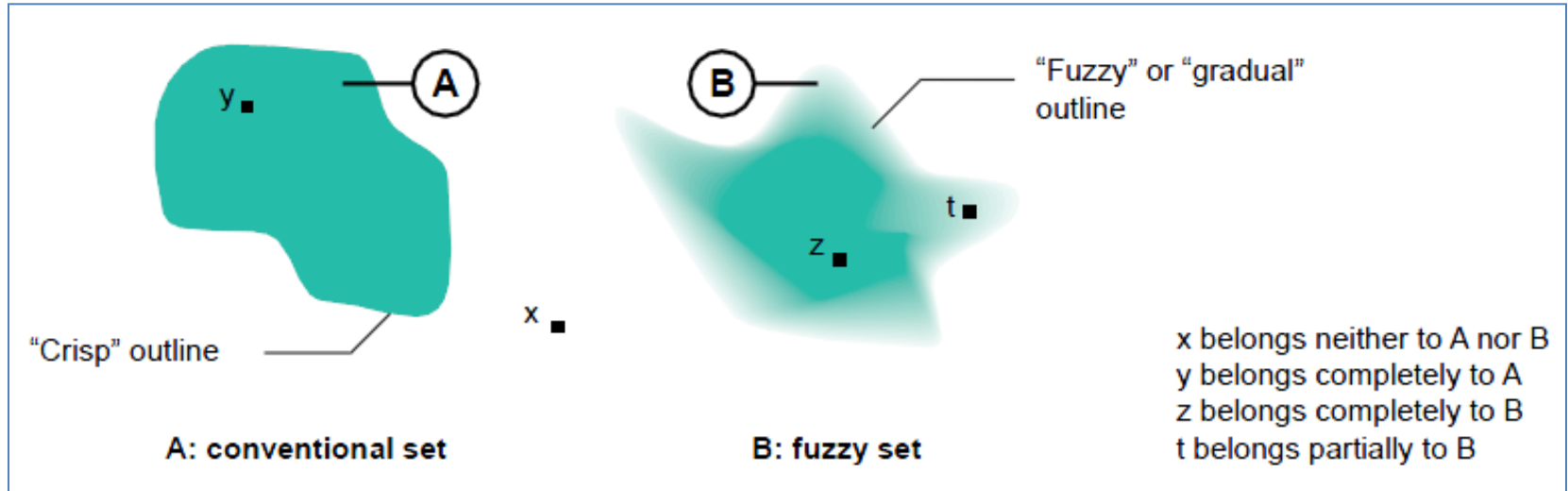
Complement

$$\bar{A} \rightarrow \chi_{\bar{A}}(x) = 1 - \chi_A(x)$$

Example

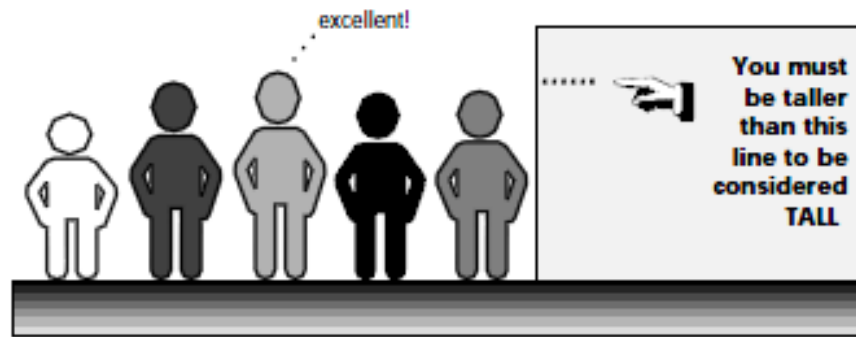
Arsha Arsh Computer Science
and Engineering, VJCET,
Vazhakulam

Fuzzy Sets



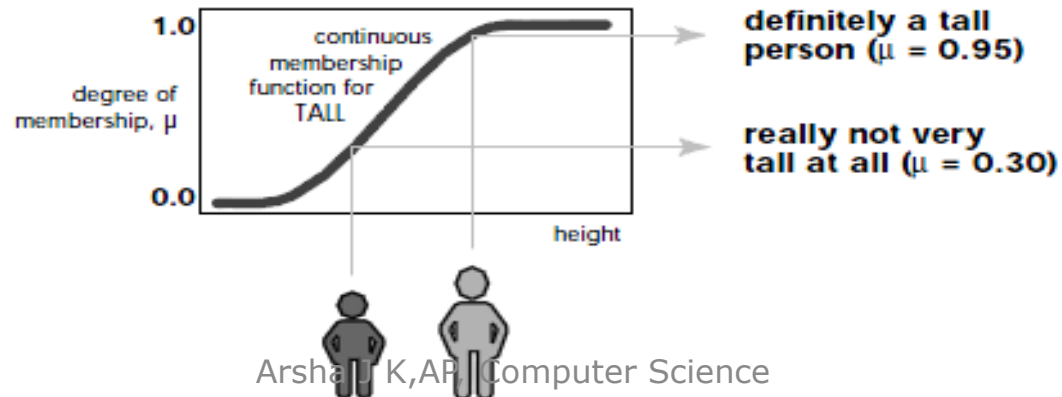
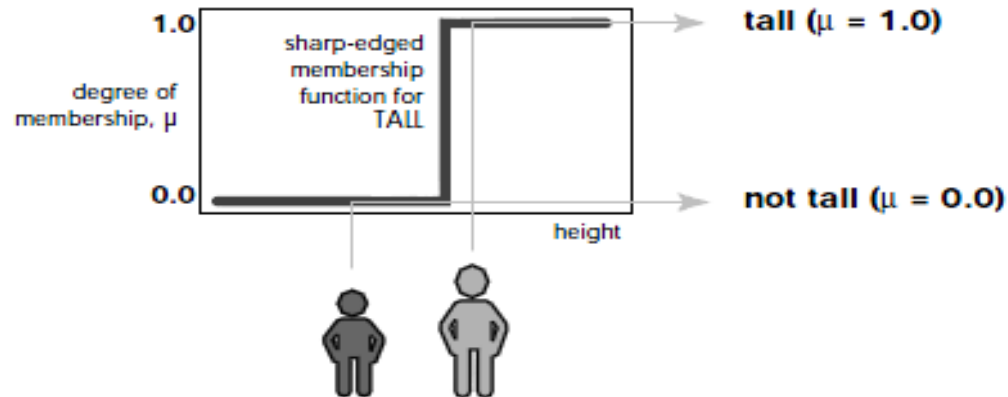
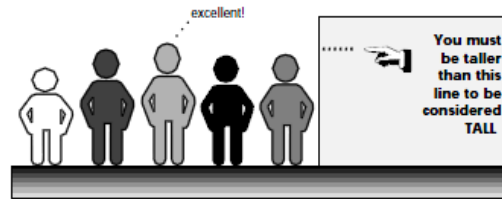
The boundaries of the fuzzy sets are vague and ambiguous. Hence, membership of an element from the universe in this set is measured by a function that attempts to describe vagueness and ambiguity

Fuzzy Sets



the set of tall people

Fuzzy Sets



Fuzzy Sets

Elements of a fuzzy set are mapped to a universe of ***membership values*** using a function-theoretic form.

fuzzy sets are denoted by a set symbol with a tilde understrike;
 $A_{\sim}$ would be the *fuzzy set A*.

This function maps elements of a fuzzy set $A_{\sim}$ to a real numbered value on the interval 0 to 1.

If an element in the universe, say x , *is a member of fuzzy set $A_{\sim}$* , then this mapping is given by

$$\mu_{A_{\sim}}(x) \in [0,1]$$

Notation Convention for Fuzzy Sets

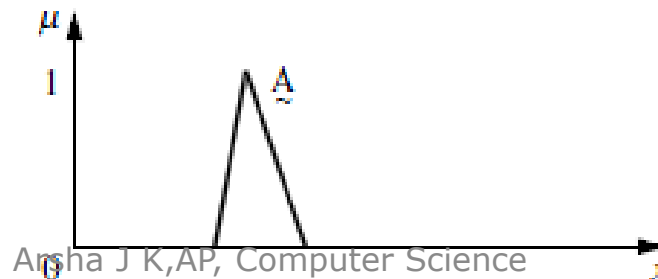
When the universe of discourse, X , is discrete and finite, is as follows for a fuzzy set $A_{\sim}$:

$$A_{\sim} = \left\{ \frac{\mu_{A_{\sim}}(x_1)}{x_1} + \frac{\mu_{A_{\sim}}(x_2)}{x_2} + \dots \right\} = \left\{ \sum_i \frac{\mu_{A_{\sim}}(x_i)}{x_i} \right\}$$

When the universe, X , is continuous and infinite, the fuzzy set $A_{\sim}$

$$A_{\sim} = \left\{ \int \frac{\mu_{A_{\sim}}(x)}{x} \right\}$$

Membership function for fuzzy set $A_{\sim}$



Fuzzy Set Operations

Three fuzzy sets A , B, and C on the universe X

For a given element x of the universe, the following **function-theoretic operations** for the set-theoretic operations of union, intersection, and complement are defined for aA, B, and C on X

Union

$$\mu_{\underline{A} \cup \underline{B}}(x) = \mu_{\underline{A}}(x) \vee \mu_{\underline{B}}(x)$$

Intersection

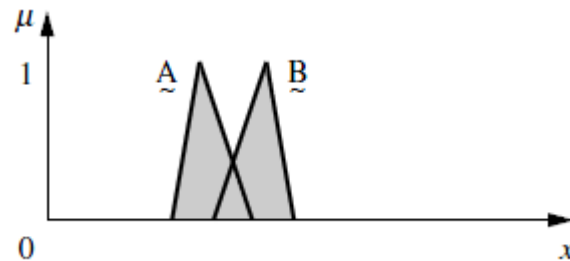
$$\mu_{\underline{A} \cap \underline{B}}(x) = \mu_{\underline{A}}(x) \wedge \mu_{\underline{B}}(x)$$

Complement

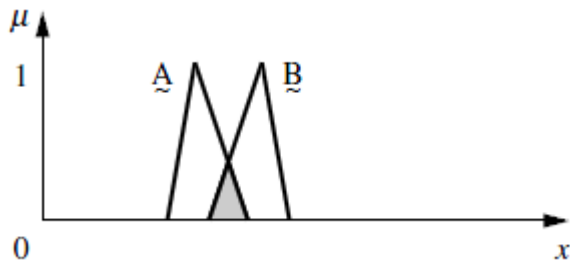
$$\mu_{\underline{A}^-}(x) = 1 - \mu_{\underline{A}}(x)$$

Standard fuzzy operations

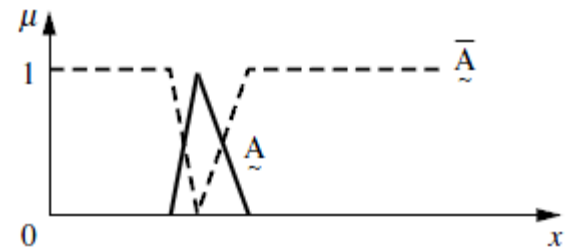
Fuzzy Set Operations



Union of fuzzy sets A and B



Intersection of fuzzy sets A and B



Complement of fuzzy sets A and B

Fuzzy Set Operations

All other operations on classical sets also hold for fuzzy sets, **except** for the **excluded middle axioms**

$$\underset{\sim}{A} \cup \overline{\underset{\sim}{A}} \neq X$$

$$\underset{\sim}{A} \cap \overline{\underset{\sim}{A}} \neq \emptyset$$

Proof, classical and fuzzy sets?

Examples of Fuzzy Set Operations

Example: chemical engineering case

Suppose the selection of an appropriate analyzer to monitor the “sales gas” sour gas concentration is important. This selection process can be complicated by the fact that one type of analyzer, say A, does not provide an average suitable pressure range but it does give a borderline value of instrument dead time; in contrast another analyzer, say B, may give a good value of process dead time but a poor pressure range.

Suppose for this problem we consider three analyzers: A, B and C.

1. the pressure range suitability of analyzers A, B, and C (a membership of 0 is not suitable, a value of 1 is excellent) ?
2. the instrument dead time suitability of analyzers A, B, and C (again, 0 is not suitable and 1 is excellent) ?
3. the analyzers that are not suitable for pressure range and instrument dead time, respectively ?
4. which analyzer is most suitable in either category ?
5. which analyzer is suitable in both categories ?

Examples of Fuzzy Set Operations

Example:

We are asked to select an **implementation technology** for a numerical processor. Computation **throughput** is directly related to **clock speed**. We are considering whether the design should be implemented using medium-scale integration (MSI) with discrete parts, field-programmable array parts (FPGA), or multichip modules (MCM).

Define the universe of potential clock speeds as MHz; and define **MSI**, **FPGA**, and **MCM** as **fuzzy sets of clock frequencies** that should be implemented in each of these technologies. The following table defines the membership values for each of the three fuzzy sets.

Clock frequency, MHz	MSI	FPGA	MCM
1	1	0.3	0
10	0.7	1	0
20	0.4	1	0.5
40	0	0.5	0.7
80	0	0.2	1
100	0	0	1

Representing the three sets as $MSI = M_{\sim}$, $FPGA = F_{\sim}$, and $MCM = C_{\sim}$, find the following:

$$M \cup F$$

$$M \cap F$$

$$M \cap C$$

Examples of Fuzzy Set Operations

Example:

Samples of a new microprocessor IC chip are to be sent to several customers for beta testing. The chips are sorted to meet certain maximum electrical characteristics, say **frequency** and **temperature** rating, so that the “best” chips are distributed to preferred customer 1. Suppose that each sample chip is screened and all chips are found to have a **maximum operating frequency** in the range 7–15 MHz at 20°C. Also, the **maximum operating temperature** range ($20^{\circ}\text{C} \pm T$) at 8 MHz is determined. Suppose there are eight sample chips with the following electrical characteristics:

	Chip number							
	1	2	3	4	5	6	7	8
$f_{\max}$, MHz	6	7	8	9	10	11	12	13
$\Delta T_{\max}$, °C	0	0	20	40	30	50	40	60

The following fuzzy sets are defined:

$\underline{A}$ = set of “fast” chips = chips with $f_{\max} \geq 12$ MHz

$\underline{B}$ = set of “slow” chips = chips with $f_{\max} \geq 8$ MHz

$\underline{C}$ = set of “cold” chips = chips with $\Delta T_{\max} \geq 10^{\circ}\text{C}$

$\underline{D}$ = set of “hot” chips = chips with $\Delta T_{\max} \geq 50^{\circ}\text{C}$

$$\underline{A} \cup \underline{D}$$

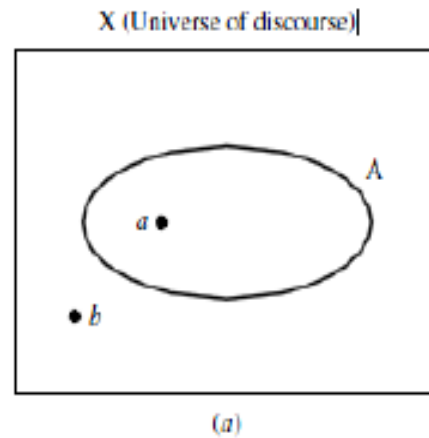
$$\underline{A} \cap \underline{D}$$

$$\overline{\underline{A} \cup \underline{D}}$$

$$\overline{\underline{A} \cap \underline{D}}$$

?

Classical Sets



the universe of discourse is the universe of all available information on a given problem
a **universe of discourse**, X , as a collection of objects all having the same characteristics

The clock speeds of computer CPUs

The operating currents of an electronic motor

The operating temperature of a heat pump (in degrees Celsius)

The Richter magnitudes of an earthquake

The integers 1 to 10

The individual elements in the universe X will be denoted as x . *The features of the elements in X can be discrete, countable integers or continuous valued quantities on the real line.*

Arsha J K, AP, Computer Science
and Engineering, VJCET,
Vezhakulam

The total number of elements in a universe X is called its **cardinal number**, denoted n_x

- The set with no element is defined as an **empty set** or null set. It is denoted by Φ
- The occurrence of an impossible event is denoted by null set, and the occurrence of the certain event indicate a whole set
- The set which consist of all possible subset of a given set A is called **power set** and is denoeted by

$$P(A) = \{x \mid x \leq A\}$$

Classical Sets

We have a universe comprised of three elements, $X = \{a, b, c\}$

The cardinal number, n_x ?

The power set, $P(X)$?

The cardinality of the power set?

OPERATIONS ON CRISP SETS

- UNION: $A \cup B = \{x | x \in A \text{ or } x \in B\}$
- INTERSECTION: $A \cap B = \{x | x \in A \text{ and } x \in B\}$
- COMPLEMENT: $\bar{A} = \{x | x \notin A, x \in X\}$
- DIFFERENCE: $A \setminus B \text{ or } (A - B) = \{x | x \in A \text{ and } x \notin B\}$
 $= A - (A \cap B)$

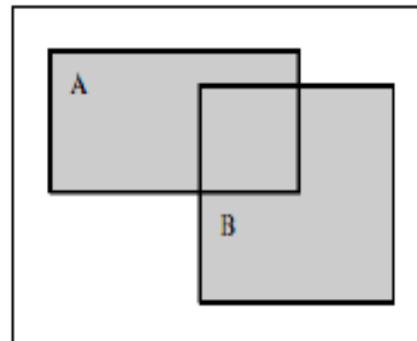
Operation on Classical Sets

Union

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

The union between the two sets, denoted $A \cup B$, represents all those elements in the universe that reside in (or belong to) the set A, the set B, or both sets A and B.

This operation is also called the **logical or**



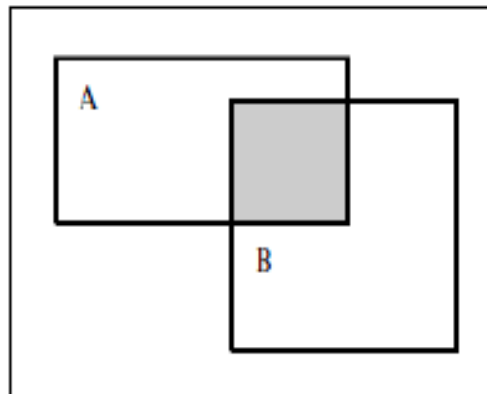
Union of sets A and B (logical or) in terms of Venn diagrams

Operation on Classical Sets

Intersection $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

The intersection of the two sets, denoted $A \cap B$, represents all those elements in the universe X that simultaneously reside in (or belong to) both sets A and B .

This operation is also called the *logical and*

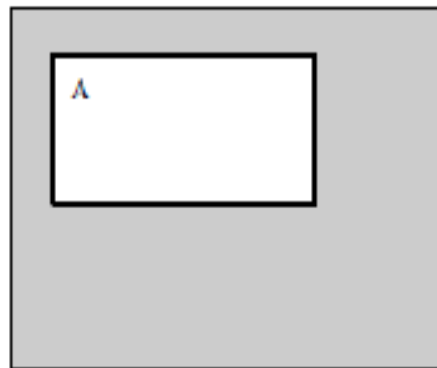


Operation on Classical Sets

Complement

$$\bar{A} = \{x \mid x \notin A, x \in X\}$$

The complement of a set A, is defined as the collection of all elements in the universe that do not reside in the set A.

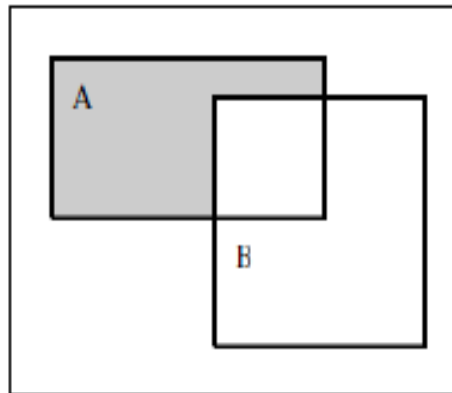


Operation on Classical Sets

Difference

$$A \mid B = \{x \mid x \in A \text{ and } x \notin B\}$$

The difference of a set A with respect to B , denoted $A \mid B$, is defined as the collection of all elements in the universe that reside in A and that do not reside in B simultaneously



Difference operation $A \mid B$

Arsha J K, AP, Computer Science
and Engineering, VJCET,
Vazhakulam

PROPERTIES OF CRISP SETS

The various properties of crisp sets are as follows:

1. Commutativity

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

2. Associativity

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

3. Distributivity

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

4. Idempotency

$$A \cup A = A$$

$$A \cap A = A$$

5. Transitivity

$$\text{If } A \subseteq B \subseteq C, \text{ then } A \subseteq C$$

6. Identity

$$\begin{aligned} A \cup \phi &= A, & A \cap \phi &= \phi \\ A \cap X &= A, & A \cup X &= X \end{aligned}$$

7. Involution (double negation)

$$\bar{\bar{A}} = A$$

8. Law of excluded middle

$$A \cup \bar{A} = X$$

9. Law of contradiction

$$A \cap \bar{A} = \phi$$

10. DeMorgan's law

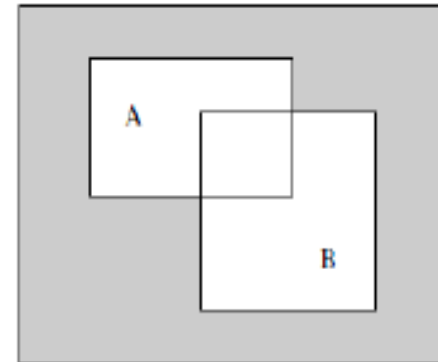
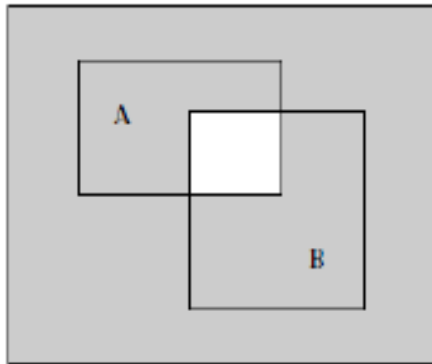
$$\begin{aligned} \overline{A \cap B} &= \bar{A} \cup \bar{B} \\ \overline{A \cup B} &= \bar{A} \cap \bar{B} \end{aligned}$$

Properties of Classical (Crisp) Sets

De Morgan's principles

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$



information about the complement of a set (or event), or the complement of combinations of sets (or events), rather than information about the sets themselves

Mapping of Classical Sets to Functions

Mapping is an important concept in relating set-theoretic forms to function-theoretic representations of information.

In its most general form it can be used to map elements or subsets on one universe of discourse to elements or sets in another universe.

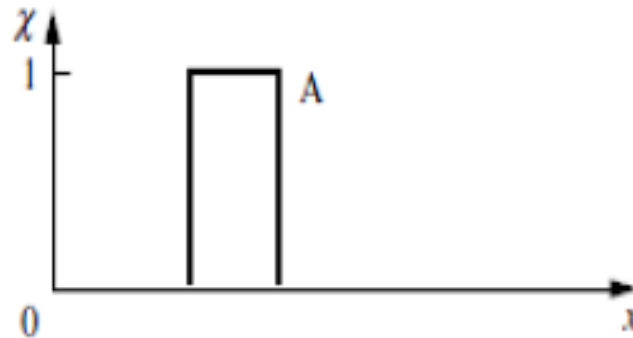
If an element x is contained in X and corresponds to an element y contained in Y , it is generally termed a mapping from X to Y ,

$$f: X \rightarrow Y$$

Mapping of Classical Sets to Functions

The characteristic (indicator) function χ_A is defined by

$$\chi_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$



Membership function is a mapping for crisp set A .

Mapping of Classical Sets to Functions

Example : a universe with three elements, $X = \{a, b, c\}$, we desire to map the elements of the power set of X , i.e., $P(X)$, to a universe, Y , consisting of only two elements (the characteristic function), $Y = \{0, 1\}$

the elements of the power set?

the elements in the value set $V(P(X))$?

Mapping of Classical Sets to Functions

Example : a universe with three elements, $X = \{a, b, c\}$, we desire to map the elements of the power set of X , i.e., $P(X)$, to a universe, Y , consisting of only two elements (the characteristic function), $Y = \{0, 1\}$

the elements of the power set

$$P(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$$

the elements in the value set $V(P(X))$

$$V\{P(X)\} = \{\{0, 0, 0\}, \{1, 0, 0\}, \{0, 1, 0\}, \{0, 0, 1\}, \{1, 1, 0\}, \{0, 1, 1\}, \{1, 0, 1\}, \{1, 1, 1\}\}$$

Mapping of Classical Sets to Functions

The union of these two sets in terms of function-theoretic terms is given as follows (the symbol $\vee$ is the maximum):

$$\text{Union } A \cup B \rightarrow \chi_{A \cup B}(x) = \chi_A(x) \vee \chi_B(x) = \max(\chi_A(x), \chi_B(x))$$

The intersection of these two sets in function-theoretic terms is given by (the symbol $\wedge$ is the minimum operator):

$$\text{Intersection } A \cap B \rightarrow \chi_{A \cap B}(x) = \chi_A(x) \wedge \chi_B(x) = \min(\chi_A(x), \chi_B(x))$$

The complement of a single set on universe X , say A , is given by

$$\text{Complement } \bar{A} \rightarrow \chi_{\bar{A}}(x) = 1 - \chi_A(x)$$

OPERATIONS ON FUZZY SETS

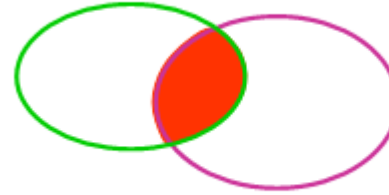
- Union: $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$
- Intersection: $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$
- Complement: $\mu_{\neg A}(x) = 1 - \mu_A(x)$

Fuzzy union operation or fuzzy OR



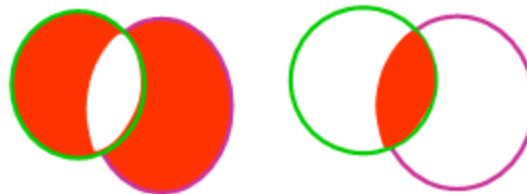
$$\mu_{A+B}(x) = \max [\mu_A(x), \mu_B(x)]$$

Fuzzy intersection operation or fuzzy AND



$$\mu_{A \cdot B}(x) = \min [\mu_A(x), \mu_B(x)]$$

Complement operation



$$\mu_{\neg A}(x) = 1 - \mu_A(x)$$

1. Commutativity

$$\begin{aligned}\tilde{A} \cup \tilde{B} &= \tilde{B} \cup \tilde{A} \\ \tilde{A} \cap \tilde{B} &= \tilde{B} \cap \tilde{A}\end{aligned}$$

2. Associativity

$$\begin{aligned}\tilde{A} \cup (\tilde{B} \cup \tilde{C}) &= (\tilde{A} \cup \tilde{B}) \cup \tilde{C} \\ \tilde{A} \cap (\tilde{B} \cap \tilde{C}) &= (\tilde{A} \cap \tilde{B}) \cap \tilde{C}\end{aligned}$$

3. Distributivity

$$\begin{aligned}\tilde{A} \cup (\tilde{B} \cap \tilde{C}) &= (\tilde{A} \cup \tilde{B}) \cap (\tilde{A} \cup \tilde{C}) \\ \tilde{A} \cap (\tilde{B} \cup \tilde{C}) &= (\tilde{A} \cap \tilde{B}) \cup (\tilde{A} \cap \tilde{C})\end{aligned}$$

4. Idempotency

$$\begin{aligned}\tilde{A} \cup \tilde{A} &= \tilde{A} \\ \tilde{A} \cap \tilde{A} &= \tilde{A}\end{aligned}$$

5. Identity

$$\begin{aligned}\tilde{A} \cup \phi &= \tilde{A} \text{ and } \tilde{A} \cup U = U (\text{universal set}) \\ \tilde{A} \cap \phi &= \phi \text{ and } \tilde{A} \cap U = \tilde{A}\end{aligned}$$

6. Involution (double negation)

$$\overline{\overline{A}} = A$$

7. Transitivity

$$\text{If } A \subseteq B \subseteq C, \text{ then } A \subseteq C$$

8. Demorgan's law

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$
$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

SUMMARY

- This chapter discussed the basic concepts of fuzzy logic.
- An introduction to crisp sets and fuzzy sets has been included.
- The operations and properties of crisp sets and fuzzy sets are also dealt with.

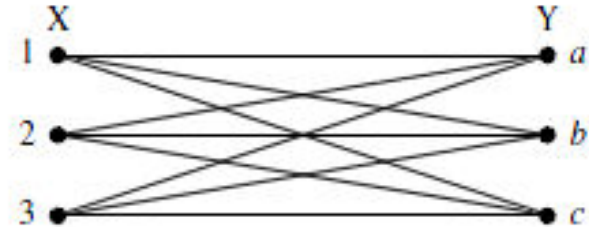
CHAPTER 8

CLASSICAL RELATIONS AND FUZZY RELATIONS

Classical Relations and Fuzzy Relations

A relation is of fundamental importance in all-engineering, science, and mathematically based fields. Relations are involved in logic, approximate reasoning, classification, rule-based systems, pattern recognition, and control.

Relations represent the *mapping* of the sets.



In the case of crisp relation there are only two degrees of relationship between the elements of sets in a crisp relation, i.e., “completely related” and “not related”. A crisp relation represents the presence or absence of association, interaction, or interconnectedness between the elements of two or more sets.

But fuzzy relations have infinite number of relationship between the extremes of completely related and not related between the elements of two or more sets considered.

RELATIONS

- Relations represent mappings between sets and connectives in logic.
- A classical binary relation represents the presence or absence of a connection or interaction or association between the elements of two sets.
- Fuzzy binary relations are a generalization of crisp binary relations, and they allow various degrees of relationship (association) between elements.

Cartesian Product

For the crisp sets $A_1, A_2, \dots, A_n$, the set of n-tuples $a_1, a_2, \dots, a_n$, where $a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n$, is called the Cartesian product of $A_1, A_2, \dots, A_n$.

The Cartesian product is denoted by $A_1 \times A_2 \times \dots \times A_n$.

In Cartesian product the first element in each pair is a member of x and the second element is a member of y formally, $X \times Y$

$$x \times y = \{(x, y) / x \in X \text{ and } y \in Y\},$$
$$\text{if } x \neq y \text{ then } x \times y \neq y \times x.$$

CRISP CARTESIAN PRODUCT

Lets consider properties of crisp relations first and then extend the mechanism to fuzzy sets.

Definition of (crisp) Product set: Let A and B be two non-empty sets, the product set or Cartesian product $A \times B$ is defined as follows,

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

(a set of ordered pairs a, b)

Cartesian Product

The elements in two sets A and B are given as

$A = \{0, 1\}$ and

$B = \{e, f, g\}$

The Cartesian product $A \times B$, $B \times A$, $A \times A$, $B \times B$?.

CRISP RELATIONS

Cartesian product of n sets

$$A_1 \times A_2 \times \dots \times A_n = \prod_{i=1}^n A_i = \{ (a_1, \dots, a_n) \mid a_1 \in A_1, \dots, a_n \in A_n \}$$

Definition of Binary Relation

If A and B are two sets and there is a specific property between elements x of A and y of B , this property can be described using the ordered pair (x, y) . A set of such (x, y) pairs, $x \in A$ and $y \in B$, is called a relation R .

$$R = \{ (x, y) \mid x \in A, y \in B \}$$

Definition of n -ary relation

For sets $A_1, A_2, A_3, \dots, A_n$, the relation among elements $x_1 \in A_1, x_2 \in A_2, x_3 \in A_3, \dots, x_n \in A_n$ can be described by n -tuple $(x_1, x_2, \dots, x_n)$. A collection of such n -tuples $(x_1, x_2, x_3, \dots, x_n)$ is a relation R among $A_1, A_2, A_3, \dots, A_n$.

$$(x_1, x_2, x_3, \dots, x_n) \in R,$$
$$R \subseteq A_1 \times A_2 \times A_3 \times \dots \times A_n$$

Classical Relations

A relation among classical sets $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ is a subset of the Cartesian product.

It is denoted by R

$$X \times Y = \{(x, y) / x \in X, y \in Y\}.$$

The strength of the relationship between ordered pairs of elements in each

universe is measured by the characteristic function denoted by χ , where a value of unity (1) is associated with complete relationship and a value of zero is associated with no relationship, i.e.,

$$\chi_{X \times Y}(x, y) = \begin{cases} 1 & (x, y) \in X \times Y, \\ 0 & (x, y) \notin X \times Y. \end{cases}$$

Classical Relations

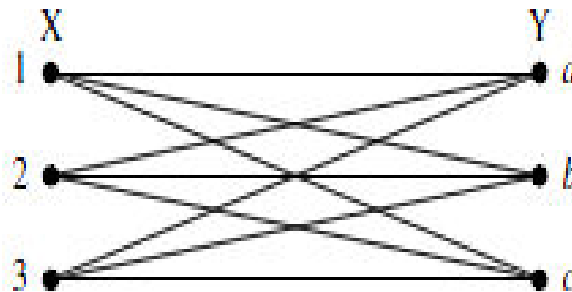
When the universe or the set are finite, a matrix can conveniently represent the relation.

The matrix is called as relation matrix

A two-dimensional matrix represents the binary relation.

If $X = \{2, 4, 6\}$ and $Y = \{p, q, r\}$, if they both are related to each other entirely, then the relation between them can be given by:

$$R = \begin{array}{ccccc} & & p & q & r \\ \begin{array}{c} 2 \\ 4 \\ 6 \end{array} & \begin{array}{c} 1 \\ 1 \\ 1 \end{array} & \begin{array}{c} 1 \\ 1 \\ 1 \end{array} & \begin{array}{c} 1 \\ 1 \\ 1 \end{array} \end{array}$$



Classical Relations

Example: Let R be a relation among the three sets

$X = \{\text{Hindi, English}\},$

$Y = \{\text{Dollar, Euro, Pound, Rupees}\}, \text{ and}$

$Z = \{\text{India, Nepal, United States, Canada}\}$

$R(x, y, z) = \{\text{Hindi, Rupees, India}\}$

$\{\text{Hindi, Rupees, Nepal}\}$

$\{\text{English, Dollar, Canada}\}$

$\{\text{English, Dollar, United States}\}.$

Relation Matrix ?

Classical Relations

Example: Let R be a relation among the three sets

$X = \{\text{Hindi, English}\}$, $Y = \{\text{Dollar, Euro, Pound, Rupees}\}$, and $Z = \{\text{India, Nepal, United States, Canada}\}$

$R(x, y, z) = \{\text{Hindi, Rupees, India}\}$
 $\{\text{Hindi, Rupees, Nepal}\}$
 $\{\text{English, Dollar, Canada}\}$
 $\{\text{English, Dollar, United States}\}.$

Solution.

	India	Nepal	US	Canada
Dollar	0	0	0	0
Euro	0	0	0	0
Pound	0	0	0	0
Rupees	1	1	0	0

Hindi

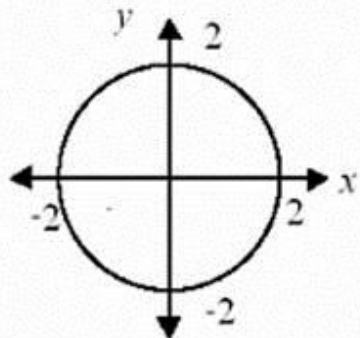
	India	Nepal	US	Canada
Dollar	0	0	1	1
Euro	0	0	0	0
Pound	0	0	0	0
Rupees	0	0	0	0

English

CRISP BINARY RELATIONS

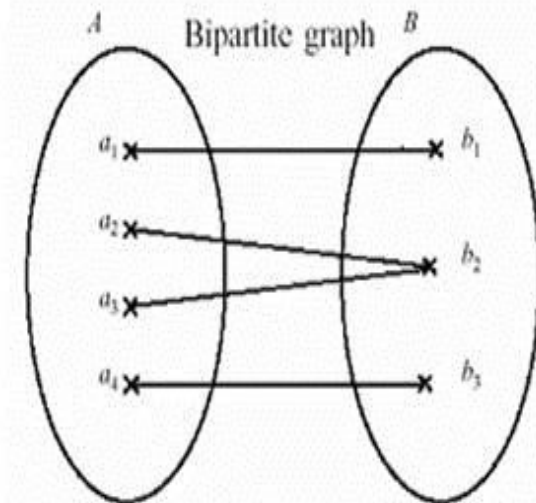
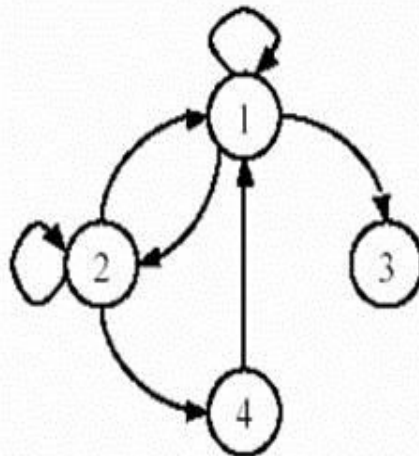
Examples of binary relations

Coordinate diagram



Relation of $x^2 + y^2 = 4$

Binary relation from A to B



Relation matrix

R	b_1	b_2	b_3
a_1	1	0	0
a_2	0	1	0
a_3	0	1	0
a_4	0	0	1

OPERATIONS ON CRISP RELATIONS

Function-theoretic operations for the two crisp relations (R, S) are defined as follows:

1. Union

$$R \cup S \rightarrow \chi_{R \cup S}(x, y) : \chi_{R \cup S}(x, y) = \max [\chi_R(x, y), \chi_S(x, y)]$$

2. Intersection

$$R \cap S \rightarrow \chi_{R \cap S}(x, y) : \chi_{R \cap S}(x, y) = \min [\chi_R(x, y), \chi_S(x, y)]$$

3. Complement

$$\overline{R} \rightarrow \chi_{\overline{R}}(x, y) : \chi_{\overline{R}}(x, y) = 1 - \chi_R(x, y)$$

4. Containment

$$R \subset S \rightarrow \chi_R(x, y) : \chi_R(x, y) \leq \chi_S(x, y)$$

5. Identity

$$\phi \rightarrow \phi_R \text{ and } X \rightarrow E_R$$

PROPERTIES OF CRISP RELATIONS

The properties of crisp sets (given below) hold good for crisp relations as well.

- Commutativity,
- Associativity,
- Distributivity,
- Involution,
- Idempotency,
- DeMorgan's Law,
- Excluded Middle Laws.

Classical Composition

Let **R** be a relation that relates, or maps, elements from universe **X** to universe **Y**, and
let **S** be a relation that relates, or maps, elements from universe **Y** to universe **Z**.

A useful question we seek to answer is whether we can find a relation, **T**, that relates the same elements in universe **X** that **R** contains to the same elements in universe **Z** that **S** contains.

we can find such a relation using an operation known as ***composition***

Classical Composition

The two methods of the composition operations are:

- Max–min composition,
- Max–product composition.

The **max–min composition** is defined by the set-theoretic and membership function-theoretic expressions:

$$T = R \circ S,$$

$$\chi_T(x, z) = \bigvee_{y \in Y} (\chi_R(x, y) \wedge \chi_S(y, z)).$$

The **max–product composition** is defined by the set-theoretic and membership function-theoretic expressions:

$$T = R \circ S,$$

$$\chi_T(x, z) = \bigvee_{y \in Y} (\chi_R(x, y) \bullet \chi_S(y, z))$$

Composition

We wish to find a **relation T** that relates the ordered pair $(x1, z2)$, *i.e.*, $(x1, z2) \in T$.
In this example,

$R = \{(x1, y1), (x1, y3), (x2, y4)\}$

$S = \{(y1, z2), (y3, z2)\}$

Sagittal diagram?

Relation matrix?

Max-min composition operations?

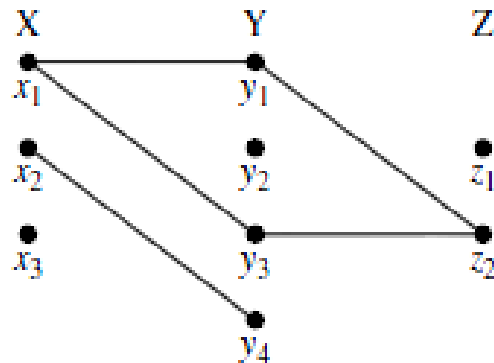
Classical Composition

We wish to find a **relation T** that relates the ordered pair $(x1, z2)$, i.e., $(x1, z2) \in T$.
In this example,

$$R = \{(x1, y1), (x1, y3), (x2, y4)\}$$

$$S = \{(y1, z2), (y3, z2)\}$$

Sagittal diagram



Classical Composition

We wish to find a **relation T** that relates the ordered pair $(x1, z2)$, i.e., $(x1, z2) \in T$.
In this example,

$$R = \{(x1, y1), (x1, y3), (x2, y4)\}$$

$$S = \{(y1, z2), (y3, z2)\}$$

Relation matrix

$$R = \begin{matrix} & \begin{matrix} y1 & y2 & y3 & y4 \end{matrix} \\ \begin{matrix} x1 \\ x2 \\ x3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad \text{and} \quad S = \begin{matrix} & \begin{matrix} z1 & z2 \end{matrix} \\ \begin{matrix} y1 \\ y2 \\ y3 \\ y4 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \end{matrix}$$

Classical Composition

We wish to find a **relation T** that relates the ordered pair $(x1, z2)$, *i.e.*, $(x1, z2) \in T$.
In this example,

$$R = \{(x1, y1), (x1, y3), (x2, y4)\}$$

$$S = \{(y1, z2), (y3, z2)\}$$

Max-Min composition operations

$$\begin{aligned}\mu_T(x_1, z_1) &= \max (\min (1, 0), \min (1, 1), \min (0, 1)) \\ &= \max [0, 1, 0] = 1,\end{aligned}$$

$$\begin{aligned}\mu_T(x_1, z_2) &= \max (\min (1, 1), \min (1, 0), \min (1, 1)) \\ &= \max [1, 0, 1] = 1,\end{aligned}$$

$$\begin{aligned}\mu_T(x_2, z_1) &= \max (\min (0, 0), \min (0, 1), \min (1, 1)) \\ &= \max [0, 0, 1] = 1,\end{aligned}$$

$$\begin{aligned}\mu_T(x_2, z_2) &= \max (\min (0, 1), \min (0, 0), \min (1, 1)) \\ &= \max [0, 0, 1] = 1,\end{aligned}$$

$$\begin{aligned}\mu_T(x_3, z_1) &= \max (\min (0, 0), \min (1, 1), \min (0, 1)) \\ &= \max [0, 1, 0] = 1,\end{aligned}$$

$$\begin{aligned}\mu_T(x_3, z_2) &= \max (\min (0, 1), \min (1, 0), \min (0, 1)) \\ &= \max [0, 0, 0] = 0,\end{aligned}$$

COMPOSITION ON CRISP RELATIONS

The composition operations are of two types:

1. Max-min composition
2. Max-product composition.

The max-min composition is defined by the function theoretic expression as

$$T = R \circ S$$
$$\chi_T(x, z) = \bigvee_{y \in Y} [\chi_R(x, y) \wedge \chi_S(y, z)]$$

The max-product composition is defined by the function theoretic expression as

$$T = R \circ S$$
$$\chi_T(x, z) = \bigvee_{y \in Y} [\chi_R(x, y) \cdot \chi_S(y, z)]$$

FUZZY CARTESIAN PRODUCT

Let R be a fuzzy subset of M and S be a fuzzy subset of N . Then the Cartesian product $R \times S$ is a fuzzy subset of $N \times M$ such that

$$\forall \vec{x} \in M, \vec{y} \in N \quad \mu_{R \times S}(\vec{x}, \vec{y}) = \min(\mu_R(\vec{x}), \mu_S(\vec{y}))$$

Example:

Let R be a fuzzy subset of $\{a, b, c\}$ such that $R = a/1 + b/0.8 + c/0.2$ and S be a fuzzy subset of $\{1, 2, 3\}$ such that $S = 1/1 + 3/0.8 + 2/0.5$. Then $R \times S$ is given by

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 1 & 0.5 & 0.9 \\ 0.8 & 0.5 & 0.8 \\ 0.2 & 0.2 & 0.2 \end{bmatrix} \end{matrix}$$

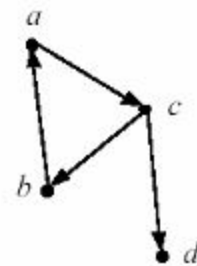
FUZZY RELATION

A fuzzy relation R is a mapping from the Cartesian space $X \times Y$ to the interval $[0,1]$, where the strength of the mapping is expressed by the membership function of the relation $\mu_R(x,y)$

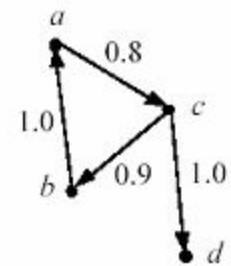
$$\mu_R : A \times B \rightarrow [0, 1]$$

$$R = \{((x, y), \mu_R(x, y)) \mid \mu_R(x, y) \geq 0, x \in A, y \in B\}$$

Crisp relation vs. Fuzzy relation



(a) Crisp relation



(b) Fuzzy relation

Corresponding fuzzy relation matrix

$A \backslash A$	a	b	c	d
a	0.0	0.0	0.8	0.0
b	1.0	0.0	0.0	0.0
c	0.0	0.9	0.0	1.0
d	0.0	0.0	0.0	0.0

OPERATIONS ON FUZZY RELATION

The basic operation on fuzzy sets also apply on fuzzy relations.

1. Union:

$$\mu_{\tilde{R} \cup \tilde{S}}(x, y) = \max [\mu_{\tilde{R}}(x, y), \mu_{\tilde{S}}(x, y)]$$

2. Intersection:

$$\mu_{\tilde{R} \cap \tilde{S}}(x, y) = \min [\mu_{\tilde{R}}(x, y), \mu_{\tilde{S}}(x, y)]$$

3. Complement:

$$\mu_{\tilde{\tilde{R}}}(x, y) = 1 - \mu_{\tilde{R}}(x, y)$$

4. Containment:

$$\tilde{R} \subset \tilde{S} \Rightarrow \mu_{\tilde{R}}(x, y) \leq \mu_{\tilde{S}}(x, y)$$

5. Inverse:

The inverse of a fuzzy relation R on $X \times Y$ is denoted by R^{-1} . It is a relation on $Y \times X$ defined by $R^{-1}(y, x) = R(x, y)$ for all pairs $(y, x) \in Y \times X$.

6. Projection:

For a fuzzy relation $R(X, Y)$, let $[R \downarrow Y]$ denote the projection of R onto Y . Then $[R \downarrow Y]$ is a fuzzy relation in Y whose membership function is defined by

$$\mu_{[R \downarrow Y]}(x, y) = \max_x \mu_R(x, y)$$

The projection concept can be extended to an n -ary relation $R(x_1, x_2, \dots, x_n)$.

PROPERTIES OF FUZZY RELATIONS

The properties of fuzzy sets (given below) hold good for fuzzy relations as well.

- Commutativity,
- Associativity,
- Distributivity,
- Involution,
- Idempotency,
- DeMorgan's Law,
- Excluded Middle Laws.

COMPOSITION OF FUZZY RELATIONS

Two fuzzy relations R and S are defined on sets A , B and C . That is, $R \subseteq A \times B$, $S \subseteq B \times C$. The composition $S \bullet R = SR$ of two relations R and S is expressed by the relation from A to C :

For $(x, y) \in A \times B$, $(y, z) \in B \times C$,

$$\mu_{S \bullet R}(x, z) = \max_y [\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_y [\mu_R(x, y) \wedge \mu_S(y, z)]$$

$M_{S \bullet R} = M_R \bullet M_S$ (matrix notation)
(max-min composition)

Example:

$$X = \{x_1, x_2\}, Y = \{y_1, y_2\}, \text{ and } Z = \{z_1, z_2, z_3\}$$

Consider the following fuzzy relations:

$$\tilde{R} = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \end{matrix} \quad \text{and} \quad \tilde{S} = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.9 & 0.6 & 0.5 \\ 0.1 & 0.7 & 0.5 \end{bmatrix} \end{matrix}$$

Using max-min composition,

$$\left. \begin{aligned} \mu_{\tilde{T}}(x_1, z_1) &= \bigvee_{y \in Y} (\mu_{\tilde{R}}(x_1, y) \wedge \mu_{\tilde{S}}(y, z_1)) \\ &= \max[\min(0.7, 0.9), \min(0.5, 0.1)] \\ &= 0.7 \end{aligned} \right\} \tilde{T} = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.6 & 0.5 \\ 0.8 & 0.6 & 0.4 \end{bmatrix} \end{matrix}$$

Two fuzzy relations R and S are defined on sets A , B and C . That is, $R \subseteq A \times B$, $S \subseteq B \times C$. The composition $S \bullet R = SR$ of two relations R and S is expressed by the relation from A to C :

For $(x, y) \in A \times B$, $(y, z) \in B \times C$,

$$\begin{aligned}\mu_{S \bullet R}(x, z) &= \max_y [\mu_R(x, y) \bullet \mu_S(y, z)] \\ &= \vee_y [\mu_R(x, y) \bullet \mu_S(y, z)]\end{aligned}$$

$M_{S \bullet R} = M_R \bullet M_S$ (matrix notation)
(max-product composition)

Max-product example:

$$X = \{x_1, x_2\}, \quad Y = \{y_1, y_2\}, \quad \text{and} \quad Z = \{z_1, z_2, z_3\}$$

Consider the following fuzzy relations:

$$\tilde{R} = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \end{matrix} \quad \text{and} \quad \tilde{S} = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.9 & 0.6 & 0.5 \\ 0.1 & 0.7 & 0.5 \end{bmatrix} \end{matrix}$$

Using max-product composition,

$$\left. \begin{aligned} \mu_{\tilde{T}}(x_2, z_2) &= \bigvee_{y \in Y} (\mu_{\tilde{R}}(x_2, y) \circ \mu_{\tilde{S}}(y, z_2)) \\ &= \max[(0.8, 0.6), (0.4, 0.7)] \\ &= 0.48 \end{aligned} \right\} \tilde{T} = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} .63 & .42 & .25 \\ .72 & .48 & .20 \end{bmatrix} \end{matrix}$$

CLASSICAL EQUIVALENCE RELATION

Let relation R on universe X be a relation from X to X . Relation R is an equivalence relation if the following three properties are satisfied.

1. Reflexivity
2. Symmetry
3. Transitivity

The function theoretic forms of representation of these properties are as follows:

1. Reflexivity:

$$\chi_R(x_i, x_i) = 1 \text{ or } (x_i, x_i) \in R$$

2. Symmetry:

$$\begin{aligned}\chi_R(x_i, x_j) &= \chi_R(x_j, x_i) \\ \text{i.e., } (x_i, x_j) \in R &\Rightarrow (x_j, x_i) \in R\end{aligned}$$

3. Transitivity:

$$\begin{aligned}\chi_R(x_i, x_j) \text{ and } \chi_R(x_j, x_k) &= 1, \text{ so } \chi_R(x_i, x_k) = 1 \\ \text{i.e., } (x_i, x_j) \in R \text{ and } (x_j, x_k) \in R &\Rightarrow (x_i, x_k) \in R\end{aligned}$$

CLASSICAL TOLERANCE RELATION

A tolerance relation R_1 on universe X is one where the only the properties of reflexivity and symmetry are satisfied. The tolerance relation can also be called proximity relation. An equivalence relation can be formed from tolerance relation R_1 by $(n - 1)$ compositions within itself, where n is the cardinality of the set that defines R_1 , here it is X , i.e.

$$\underbrace{R_1^{n-1}}_{\substack{\text{Tolerance} \\ \text{relation}}} = R_1 \circ R_1 \circ \cdots \circ R_1 = \underbrace{R}_{\substack{\text{Equivalence} \\ \text{relation}}}$$

- Suppose we have a tolerance relation

- $R_1 =$

1	1	0	0	0
1	1	0	0	1
0	0	1	0	0
0	0	0	1	0
0	1	0	0	1

- This relation become an equivalence relation by one composition

- $R = R1 \cdot R1 =$

1	1	0	0	1
1	1	0	0	1
0	0	1	0	0
0	0	0	1	0
1	1	0	0	1

FUZZY EQUIVALENCE RELATION

Let $\tilde{R}$ be a fuzzy relation on universe X , which maps elements from X to X . Relation $\tilde{R}$ will be a fuzzy equivalence relation if all the three properties – reflexive, symmetry and transitivity – are satisfied. The membership function theoretic forms for these properties are represented as follows:

1. Reflexivity:

$$\mu_{\tilde{R}}(x_i, x_i) = 1 \quad \forall x \in X$$

If this is not the case for few $x \in X$, then $R(X, X)$ is said to be irreflexive.

2. Symmetry:

$$\mu_{\tilde{R}}(x_i, x_j) = \mu_{\tilde{R}}(x_j, x_i) \text{ for all } x_i, x_j \in X$$

If this is not satisfied for few $x_i, x_j \in X$, then $R(X, X)$ is called asymmetric.

3. Transitivity:

$$\mu_{\tilde{R}}(x_i, x_j) = \lambda_1 \text{ and } \mu_{\tilde{R}}(x_j, x_k) = \lambda_2$$

$$\Rightarrow \mu_{\tilde{R}}(x_i, x_k) = \lambda$$

where

$$\lambda = \min [\lambda_1, \lambda_2]$$

$$\text{i.e., } \mu_{\tilde{R}}(x_i, x_j) \geq \max_{x_j \in X} \min[\mu_{\tilde{R}}(x_i, x_j), \mu_{\tilde{R}}(x_j, x_k)] \quad \forall (x_i, x_k) \in X^2$$

FUZZY TOLERANCE RELATION

A binary fuzzy relation that possesses the properties of reflexivity and symmetry is called fuzzy tolerance relation or resemblance relation.

The equivalence relations are a special case of the tolerance relation. The fuzzy tolerance relation can be reformed into fuzzy equivalence relation in the same way as a crisp tolerance relation is reformed into crisp equivalence relation, i.e.,

$$\underbrace{R_1^{n-1}}_{\substack{\text{Fuzzy} \\ \text{tolerance} \\ \text{relation}}} = R_1 \circ R_1 \circ \cdots \circ R_1 = \underbrace{R}_{\substack{\text{Fuzzy} \\ \text{equivalence} \\ \text{relation}}}$$

where 'n' is the cardinality of the set that defines R1.

SUMMARY

The chapter provides the definition, properties, and operations of the following:

- Classical sets,
- Fuzzy sets,
- Classical relations,
- Fuzzy relations.

The chapter also discusses crisp and fuzzy equivalence and tolerance relations.

Fuzzy Relation

Fuzzy relations are fuzzy subsets of $X \times Y$, i.e., mapping from $X \rightarrow Y$.

Fuzzy relations maps elements of one universe, X to those of another universe, say Y , through the Cartesian product of the two universes.

A fuzzy relation $R_{\sim}$ is mapping from the Cartesian space $X \times Y$ to the interval $[0, 1]$ where the strength of the mapping is expressed by the **membership function of the relation** for ordered pairs. This can be expressed as

$$R_{\sim} = \{ ((x, y), \mu_{R_{\sim}}(x, y)) \mid (x, y) \in X \times Y \}$$

is called a fuzzy relation on $X \times Y$.

Fuzzy Relation

Cartesian Product

Let $A_{\sim}$ be a fuzzy set on universe X and $B_{\sim}$ be a fuzzy set on universe Y , then the Cartesian product between fuzzy sets $A_{\sim}$ and $B_{\sim}$ will result in a fuzzy relation $R_{\sim}$ which is contained with the full Cartesian product space or

$$A_{\sim} \times B_{\sim} = R_{\sim} \subset X \times Y,$$

where the fuzzy relation $R_{\sim}$ has membership function.

$$\mu_{R_{\sim}}(x, y) = \mu_{A_{\sim} \times B_{\sim}}(x, y) = \min \left(\mu_{A_{\sim}}(x), \mu_{B_{\sim}}(y) \right)$$

Fuzzy Relation

Cartesian Product

Example: Suppose we have two fuzzy sets, $A_{\sim}$ defined on a universe of three discrete temperatures, $X = \{x_1, x_2, x_3\}$, and $B_{\sim}$ defined on a universe of two discrete pressures, $Y = \{y_1, y_2\}$, and we want to find the fuzzy Cartesian product between them. Fuzzy set $A_{\sim}$ could represent the “ambient” temperature and fuzzy set $B_{\sim}$ the “near optimum” pressure for a certain heat exchanger, and the Cartesian product might represent the conditions (temperature–pressure pairs) of the exchanger that are associated with “efficient” operations. For example, let

$$A_{\sim} = \frac{0.2}{x_1} + \frac{0.5}{x_2} + \frac{1}{x_3} \quad \text{and} \quad B_{\sim} = \frac{0.3}{y_1} + \frac{0.9}{y_2}$$

Fuzzy Relation

Cartesian Product

Example: Suppose we have two fuzzy sets, $A_{\sim}$ defined on a universe of three discrete temperatures, $X = \{x_1, x_2, x_3\}$, and $B_{\sim}$ defined on a universe of two discrete pressures, $Y = \{y_1, y_2\}$, and we want to find the fuzzy Cartesian product between them. Fuzzy set $A_{\sim}$ could represent the “ambient” temperature and fuzzy set $B_{\sim}$ the “near optimum” pressure for a certain heat exchanger, and the Cartesian product might represent the conditions (temperature–pressure pairs) of the exchanger that are associated with “efficient” operations. For example, let

$$A_{\sim} = \frac{0.2}{x_1} + \frac{0.5}{x_2} + \frac{1}{x_3} \quad \text{and} \quad B_{\sim} = \frac{0.3}{y_1} + \frac{0.9}{y_2}$$

The fuzzy Cartesian product results in a **fuzzy relation** $R_{\sim}$ representing “**efficient**” conditions,

$$A_{\sim} \times B_{\sim} = R_{\sim} = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.2 & 0.2 \\ 0.3 & 0.5 \\ 0.3 & 0.9 \end{bmatrix} \end{matrix}$$

Fuzzy Relation Composition

Fuzzy composition can be defined just as it is for crisp (binary) relations. Suppose $\underline{R}$ is a fuzzy relation on the Cartesian space $X \times Y$, $\underline{S}$ is a fuzzy relation on $Y \times Z$, and $\underline{T}$ is a fuzzy relation on $X \times Z$;

then **fuzzy max–min composition** is defined in terms of the **set-theoretic notation** and **membership function-theoretic notation** in the following manner:

$$\underline{T} = \underline{R} \circ \underline{S}$$

$$\mu_{\underline{T}}(x, z) = \bigvee_{y \in Y} (\mu_{\underline{R}}(x, y) \wedge \mu_{\underline{S}}(y, z))$$

fuzzy max–product composition is defined in terms of the **membership function theoretic notation** as

$$\mu_{\underline{T}}(x, z) = \bigvee_{y \in Y} (\mu_{\underline{R}}(x, y) \bullet \mu_{\underline{S}}(y, z))$$

Fuzzy Relation Composition

Example: Fuzzy relationships
for $X \times Y$ (denoted by the fuzzy relation $R_{\sim}$)
and $Y \times Z$ (denoted by the fuzzy relation $S_{\sim}$).

$X = \{x_1, x_2\}$, $Y = \{y_1, y_2\}$, and $Z = \{z_1, z_2, z_3\}$

Consider the following fuzzy relations:

$$R_{\sim} = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.6 \\ 0.8 & 0.3 \end{bmatrix} \end{matrix}, \quad S_{\sim} = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.8 & 0.5 & 0.4 \\ 0.1 & 0.6 & 0.7 \end{bmatrix} \end{matrix}$$

Find the resulting relation, $T_{\sim}$, which relates elements of universe X to elements of universe Z , i.e., defined on Cartesian space $X \times Z$
using **max-min** composition and
using **max-product** composition

Fuzzy Relation Composition

Solution:Max–Min Composition

$$R = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.6 \\ 0.8 & 0.3 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.8 & 0.5 & 0.4 \\ 0.1 & 0.6 & 0.7 \end{bmatrix} \end{matrix}$$

$$T = R \circ S$$

$$\begin{aligned} \mu_T(x_1, z_1) &= \max [\min (0.7, 0.8), \min (0.6, 0.1)] \\ &= \max [0.7, 0.1] \\ &= 0.7, \end{aligned}$$

$$\begin{aligned} \mu_T(x_1, z_2) &= \max [\min (0.7, 0.5), \min (0.6, 0.6)] \\ &= \max [0.5, 0.6] \\ &= 0.6, \end{aligned}$$

$$\begin{aligned} \mu_T(x_1, z_3) &= \max [\min (0.7, 0.4), \min (0.6, 0.7)] \\ &= \max [0.4, 0.7] \\ &= 0.7, \end{aligned}$$

$$\begin{aligned} \mu_T(x_2, z_1) &= \max [\min (0.8, 0.8), \min (0.3, 0.1)] \\ &= \max [0.8, 0.1] \\ &= 0.8, \end{aligned}$$

$$\begin{aligned} \mu_T(x_2, z_2) &= \max [\min (0.8, 0.5), \min (0.3, 0.6)] \\ &= \max [0.5, 0.3] \\ &= 0.5, \end{aligned}$$

$$\begin{aligned} \mu_T(x_2, z_3) &= \max [\min (0.8, 0.4), \min (0.3, 0.7)] \\ &= 0.4, \end{aligned}$$

$$T = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.6 & 0.7 \\ 0.8 & 0.5 & 0.4 \end{bmatrix} \end{matrix}$$

Fuzzy Relation Composition

Solution:Max–Product Composition

$$R = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.6 \\ 0.8 & 0.3 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.8 & 0.5 & 0.4 \\ 0.1 & 0.6 & 0.7 \end{bmatrix} \end{matrix}$$

$$\begin{aligned} \mu_T(x_1, z_1) &= \max [\min (0.7 \times 0.8), \min (0.6 \times 0.1)] \\ &= \max [0.56, 0.06] \\ &= 0.56, \end{aligned}$$

$$\begin{aligned} \mu_T(x_1, z_2) &= \max [\min (0.7 \times 0.5), \min (0.6 \times 0.6)] \\ &= \max [0.35, 0.36] \\ &= 0.36, \end{aligned}$$

$$\begin{aligned} \mu_T(x_1, z_3) &= \max [\min (0.7 \times 0.4), \min (0.6 \times 0.7)] \\ &= \max [0.28, 0.42] \\ &= 0.42, \end{aligned}$$

$$\begin{aligned} \mu_T(x_2, z_1) &= \max [\min (0.8 \times 0.8), \min (0.3 \times 0.1)] \\ &= \max [0.64, 0.03] \\ &= 0.64, \end{aligned}$$

$$\begin{aligned} \mu_T(x_2, z_2) &= \max [\min (0.8 \times 0.5), \min (0.3 \times 0.6)] \\ &= \max [0.40, 0.18] \\ &= 0.40, \end{aligned}$$

$$\begin{aligned} \mu_T(x_2, z_3) &= \max [\min (0.8 \times 0.4), \min (0.3 \times 0.7)] \\ &= \max [0.32, 0.21] \\ &= 0.32, \end{aligned}$$

$$T = \begin{bmatrix} 0.56 & 0.36 & 0.42 \\ 0.64 & 0.40 & 0.32 \end{bmatrix}.$$